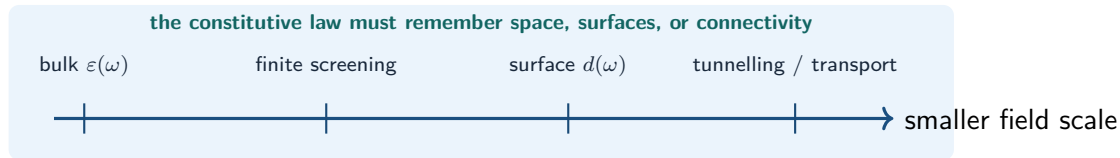


From local Maxwell to mesoscopic response

When does a nanostructure stop looking like a bulk material?



A critical map of the model hierarchy, not a chronological paper summary.

Review cutoff: 10 June 2026. Main evidence is concentrated in 2009–2025, with 1970/1982 conceptual anchors.

How to read this journal-club talk

The prescribed questions

- ▶ What is genuinely measured or computed?
- ▶ Which conclusion follows directly from that evidence?
- ▶ What assumptions make the inference possible?
- ▶ Which alternative mechanism could mimic the same signal?

The thesis of this talk

Local electrodynamics is not “wrong at the nanoscale.” It is a controlled zeroth-order model whose failure is set by the *smallest field scale*, not by the label “nano.”

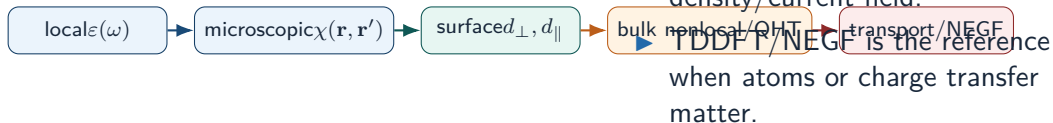
My standard for a convincing nonlocal claim

The same parameter set should account for at least a reactive observable and a dissipative or transport observable.

Central question and short answer

Scientific question

How far can a nanostructure be scaled before bulk local $\epsilon(\omega)$, a sharp interface, and classical Maxwell boundary conditions stop being predictive? Which replacement is justified: microscopic response, surface response, hydrodynamics, or quantum transport?



Short answer

These are not rival “quantum corrections.” They retain different pieces of the same two-point response and have different domains of validity.

- ▶ d compresses interface structure.
- ▶ HDM/QHT retains a continuum density/current field.

▶ PDDFT/NEGF is the reference when atoms or charge transfer matter.

Local response is a sharp-interface approximation

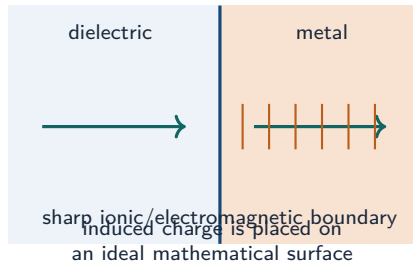
What LRA keeps

$$\mathbf{P}(\mathbf{r}, \omega) = \varepsilon_0 \chi(\omega) \mathbf{E}(\mathbf{r}, \omega),$$

$$\mathbf{J}(\mathbf{r}, \omega) = \sigma(\omega) \mathbf{E}(\mathbf{r}, \omega),$$

$$\varepsilon_m(\omega) = \varepsilon_\infty - \frac{\omega_p^2}{\omega(\omega + i\gamma)}.$$

Geometry, radiation, absorption, and bulk dispersion can be treated accurately by Mie, BEM, FEM, or FDTD.



What LRA discards

Spatial dispersion $\varepsilon(\mathbf{k}, \omega)$, diffuse equilibrium density, finite charge centroids, atomistic transitions, and tunnel conductivity.

The smallest feature controls the breakdown

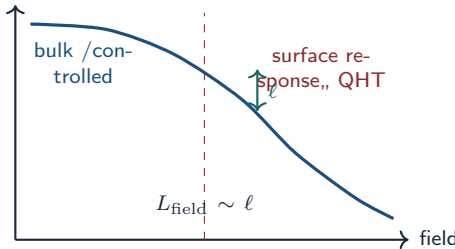
The diagnostic is a scale comparison

$$q\ell, \quad \frac{\ell}{R}, \quad \frac{\ell}{g}, \quad \frac{\ell}{t}$$

Here ℓ may be a screening length, spill-out width, Fermi wavelength, mean free path, tunnelling decay length, or exciton coherence radius.

- ▶ A 100 nm antenna with a 1 nm gap contains $q \sim 1/g$ components.
- ▶ A large isolated particle can be less nonlocal than a smaller, strongly

model confidence



What experiments actually establish

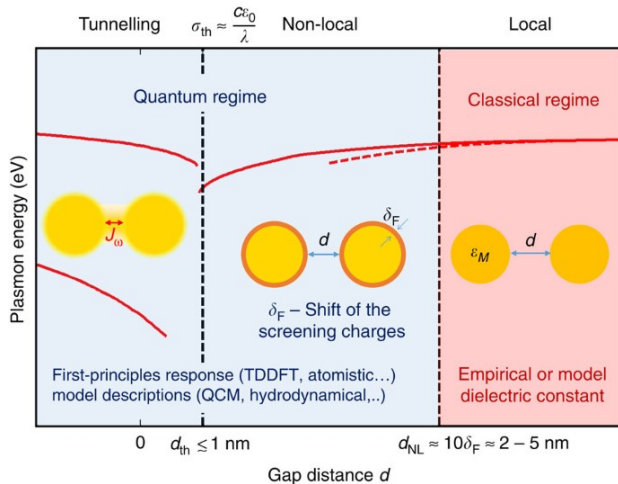
Platform	Direct observation
size-dependent EELS shifts [[?]]	few-nm Ag particles
sub-nm Au junctions Au-AIO _x cavities	optical crossover tracks conductance [[?, ?]] > 30% nonclassical shifts; non-Kreibig broadening [[?]]
2–5 nm MDM gaps Au–air interface	complex q reveals excess gap-dependent loss [[?]] broadband complex $d_{\perp}(\omega)$ retrieval [[?]]

Critical reading

A deviation from LRA is evidence for missing physics, not proof of one unique mechanism.

- ▶ A shift can be nonlocal screening, spill-out, spacer error, shape, substrate, or chemistry.
- ▶ A linewidth can be Landau damping, roughness, grains, adsorbates, radiation, or wrong bulk ε .
- ▶ The strongest evidence combines energy and linewidth,^{6 / 25}

The quantum-regime map: screening before tunnelling



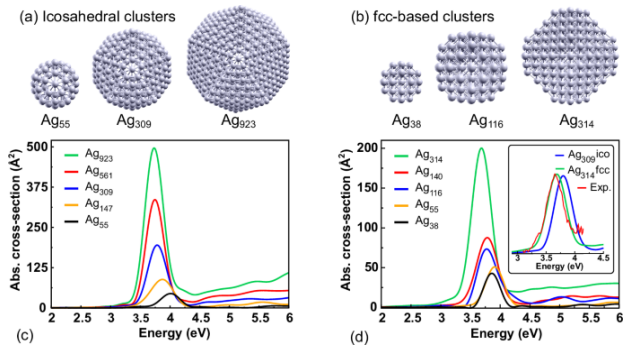
The sequence matters

1. At large d : local Maxwell with empirical ϵ_M .
2. At $d \lesssim$ a few nm: screening charge shifts by δ_F ; the effective gap differs from the geometric gap.
3. At sub-nm d : optical tunnelling current changes boundary topology; charge-transfer plasmons emerge before geometric contact.

Five mechanisms that are easy to conflate

Mechanism	LRA assumption	Observable consequence
charge can be compressed without spatial cost	longitudinal response; usually a blueshift spill-out / spill-in	finite compressibility density terminates at the ionic boundary
shifted charge centroid; red or blue shift surface Landau damping	all loss is in bulk $\epsilon(\omega)$	q - and size-dependent broadening quantum confinement
spectrum is bulk-continuous	discrete transitions, shell/subband effects tunnelling / chemistry	gap stays insulating and inert
field saturation, quenching, charge-transfer		

Microscopic benchmark: powerful, but not a device solver



Chaudhary and Weissker, *Nature Communications* 15, 9225 (2024), Fig. 2;
open-access article.

What the calculation buys

RT-TDDFT+ U retains atomic structure, d -electron screening, discrete electron-hole excitations, and the crossover to a broad LSPR.

- ▶ Ag₄ through Ag₉₂₃, roughly a 3-nm particle.
- ▶ Agreement with experiment across sizes is a real benchmark for the electronic model.
- ▶ The same scale is still far 9 / 25

Why the microscopic route remains expensive

The reference response

$$P_i(\mathbf{r}, \omega) = \varepsilon_0 \int \chi_{ij}(\mathbf{r}, \mathbf{r}'; \omega) E_j(\mathbf{r}', \omega) d\mathbf{r}'.$$

This two-point object contains surface states, tunnelling, correlations, and finite screening. The difficulty is electronic-state representation, not Maxwell propagation.

Bottleneck

variables grow
with every
design detail
occupied–
unoccupied
pairs
fine steps and
long runs for
spectral res-
olution open
boundaries

Device-level consequence

atomic surface / lig-
and / substrate
broadband response
and parameter
sweeps are costly
real-time propagation

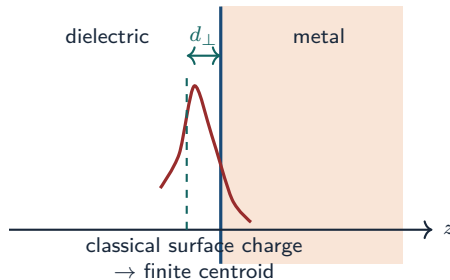
NEGF/supercells for
optical + transport
response functional
uncertainty

Feibelman parameters: compress the interface, not the bulk

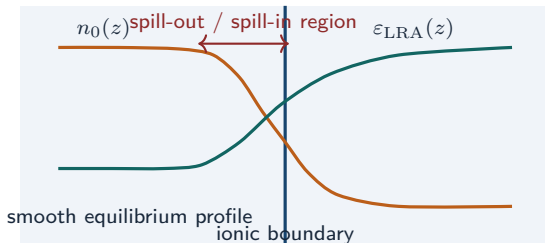
Induced-charge and current centroids

$$d_{\perp}(\omega) = \frac{\int z \rho_{\text{ind}}(z, \omega) dz}{\int \rho_{\text{ind}}(z, \omega) dz},$$
$$d_{\parallel}(\omega) = \frac{\int z \partial_z J_{\parallel}^{\text{ind}}(z, \omega) dz}{\int \partial_z J_{\parallel}^{\text{ind}}(z, \omega) dz}.$$

- ▶ $\Re d$: reactive displacement of the effective optical boundary.
- ▶ $\Im d$: surface-enabled dissipation.
- ▶ Leading effects scale as d/R or d/g .



Equilibrium inhomogeneity is not dynamical nonlocality



The 2021 conceptual result

Even with a pointwise local constitutive relation, replacing a piecewise-constant dielectric profile by a smooth profile creates a finite-width induced charge and therefore a finite d parameter.

But this is not the whole story

Finite electron-gas compressibility, Friedel oscillations, facets, and correlations add a separate dynamical contribution. Spill-out and nonlocality may reinforce or cancel.^{12 / 25}

Hydrodynamic models: what they add, what they cannot

Hard-wall HDM

$$\beta^2 \nabla(\nabla \mathbf{J}) + \omega(\omega + i\gamma) \mathbf{J} = i\omega \epsilon_0 \omega_p^2 \mathbf{E}, \quad \mathbf{J} \cdot \mathbf{n} = 0.$$

Finite compressibility produces longitudinal fields and usually a blueshift. The hard wall excludes equilibrium spill-out and tunnelling.

GNOR and SC-HDM/QHT

GNOR adds an effective diffusive damping channel. SC-HDM/QHT solves for a diffuse $n_0(\mathbf{r})$ and then linearizes dynamics around it.

Model	<u>Best defensible use</u>
	HDM
controlled long-wavelength compressibility GNOR	complex propagation / effective nonlocal loss SC-HDM
diffuse density plus continuum dynamics d response	transfer a calibrated interface into BEM/FEM QCM / NEGF
conductive gap and optical transport	

The 2026 anchor paper: a computational question

Question

Can self-consistent hydrodynamics retain a diffuse equilibrium density and its dynamical response without sacrificing the practical speed that makes continuum models attractive?

- ▶ Target geometry: homogeneous spheres, $R = 1\text{--}5\text{ nm}$.
- ▶ Material: sodium-like Drude metal, $\hbar\omega_p = 6.04\text{ eV}$.
- ▶ Functional: Thomas–Fermi + full von Weizsäcker ($\eta = 1$) + Perdew–Zunger LDA exchange–correlation.
- ▶ Unknown: whether the solver's speed survives a realistic geometry

What this paper is not

It is not a new optical measurement, not a noble-metal calibration, and not a universal $d(q, \omega)$ extraction.

What it is

A symmetry-adapted volume-integral formalism that turns the spherical SC–HDM problem into coupled radial blocks.

Why the spherical volume-integral reduction is fast



Angular separation

The sphere makes angular momentum exact rather than approximate.

No exterior volume mesh

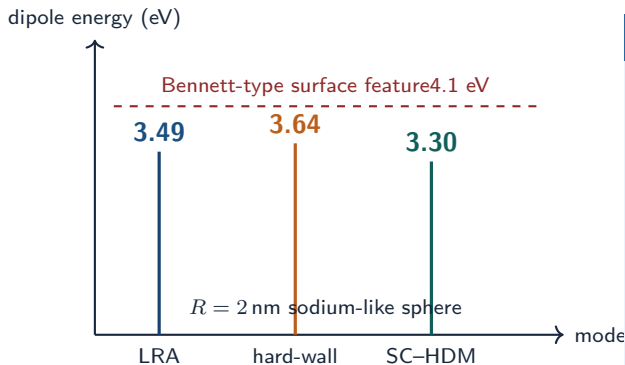
Radiation is carried by the homogeneous Green tensor.

The caveat

The speedup is symmetry-enabled; arbitrary antennas need new evidence.

Adapted from the supplied review, pp. 12–13; original anchor paper [\[\[?\]\]](#).

Anchor result: the sign of the shift diagnoses the surface model



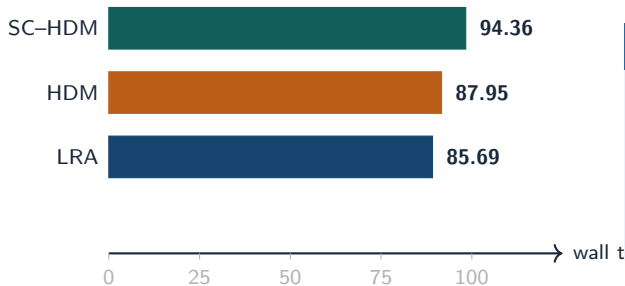
Directly reported

- ▶ LRA $\rightarrow$ 3.49 eV.
- ▶ Hard-wall HDM $\rightarrow$ 3.64 eV: compressibility pushes charge inward.
- ▶ SC-HDM $\rightarrow$ 3.30 eV: diffuse density and induced charge spill outward.
- ▶ Bennett-type feature near 4.1 eV.

Interpretive limit

The redshift is not “the nonlocal”^{16 / 25}

Runtime is a result; it is not validation of the physics



What the number supports

In this canonical spherical implementation, SC-HDM costs about 10% more than LRA. That is a meaningful computational demonstration.

What it does not support

It does not establish comparable speed for arbitrary 3D antennas, substrates, edges, touching junctions, or material-realistic interband spectra.

What the anchor paper establishes — and leaves open

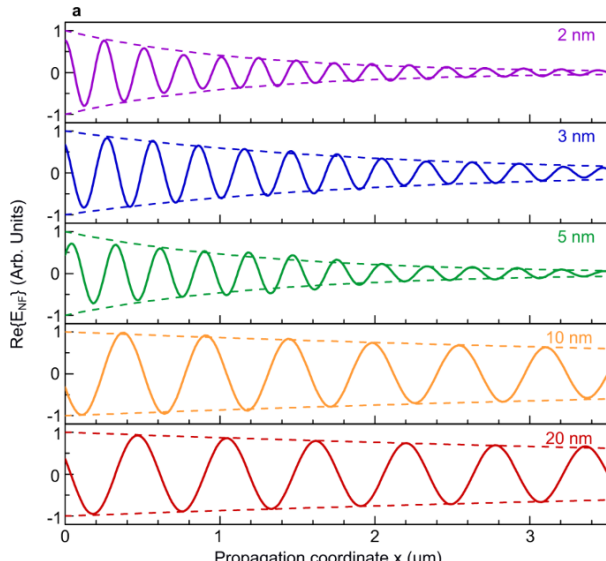
Established by the calculation

- ▶ VIE + spherical basis reproduces LRA/HDM benchmarks to about 0.08% in the reported metrics.
- ▶ A diffuse equilibrium density changes the resonance and supports a Bennett-type surface mode.
- ▶ SC-HDM can approach LRA runtime for tested spheres.
- ▶ The algebra can host alternative orbital-free functionals.

Still open

- ▶ Does the efficiency survive without spherical symmetry?
- ▶ Can the finite-sphere centroid become a causal, transferable planar $d(q, \omega)$?
- ▶ How are interband screening, facets, ligands, and substrates calibrated?
- ▶ What controls the extra high-energy zeros and features?
- ▶ Where does the continuum stop and NEGF/TDDFT become necessary?

An experiment can close the inference gap

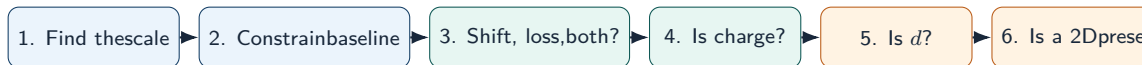


Why this is a good benchmark

s-SNOM measures the complex propagation constant of gap surface plasmons in crystalline Au-Al₂O₃-Au waveguides.

- ▶ The real part tracks confinement.
- ▶ The imaginary part tracks propagation loss.
- ▶ Few-nm gaps show excess loss relative to LRA.
- ▶ GNOR can fit the complex trend, but its diffusion

A practical model-selection workflow



Separated bodies

$d_{\perp}, d_{\parallel}$, HDM, or QHT are plausible if $d \ll g, R$ and no tunnel current flows.

Conductive junction

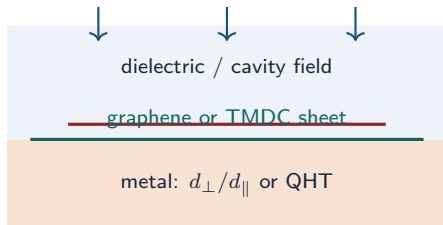
Switch to QCM calibrated by TDDFT/NEGF or explicit transport; topology has changed.

2D hybrid

Keep metal surface response and $\sigma_{2D}(q, \omega)$ or $\chi_{2D}(q, \omega)$ separate.

Workflow condensed from the supplied review, §7.4.

A metal–2D hybrid contains two nonlocal subsystems



Subsystem	Response object
$d_{\perp}(\omega), d_{\parallel}(\omega)$ or $d(q, \omega)$	metal surface $\sigma_{2D}(q, \omega)$ TMDC exciton
graphene $\chi_{2D}(q, \omega)$, often BSE/polariton informed cavity	Maxwell Green tensor and geometry

Do not hide both in one effective ε
 Otherwise momentum dependence and
 loss are assigned to the wrong subsystem,
 and the model cannot say what

Critical verdict: data, interpretation, and caveat

Evidence

a fixed bulk local ε is insufficient

GNOR diffusion is an effective fit, not a unique mechanism

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Conclusions and open questions

1. **Local Maxwell remains the zeroth-order theory.** Its validity is controlled by $q\ell$, ℓ/R , ℓ/g , and ℓ/t .
2. **Surface response and hydrodynamics are complementary reductions.** d compresses an interface; QHT keeps continuum density/current dynamics.
3. **The 2026 anchor paper solves a real computational bottleneck.** The 10% spherical overhead is persuasive as a solver result, not yet as material validation.
4. **Experiments must be overdetermined.** Pair energy with linewidth, complex propagation, conductance, or an independent microscopic benchmark.

Open boundary of the field

The angstrom conductive junction: surface polarization, atomic chemistry, and quantum transport become inseparable.

Discussion: Which observable would you add first to validate SC-HDM for a real noble-metal geometry?

Source trace: anchor-paper result page from the supplied review

MESOSCOPIC OPTICAL RESPONSE

STRUCTURED REVIEW - 2026

The paper does not report an experiment. It develops and validates a computational formalism for spherical nanoparticles and uses sodium-like material parameters, where a free-electron continuum is most credible.

6.2 Volume-integral and spherical reduction

The scattered field is written through the homogeneous electromagnetic dyadic Green tensor as a volume integral over the polarization $\mathbf{P}$. Substitution into the hydrodynamic constitutive equation yields an integral equation for $\mathbf{P}$ alone. The polarization is expanded in vector spherical harmonics and cubic radial Lagrange functions; Galerkin testing performs the angular integrals analytically.

Design choice	Consequence
Vector spherical harmonics	Exact angular-momentum separation for a sphere
TE/TM/longitudinal basis	TE decouples; TM and longitudinal sectors form small coupled blocks
Cubic radial elements	One-dimensional resolution of diffuse equilibrium and induced density
Frequency-independent material blocks	Quantum operators are assembled once across a spectrum
Volume-integral Green formulation	Radiation is handled without meshing a large exterior domain
Functional solver generator	Different orbital-free $G[n]$ closures can reuse the same algebraic framework

The speedup is therefore symmetry-enabled. It should not be generalized to arbitrary three-dimensional antennas without demonstrating a comparable reduction.

6.3 Validation and physical results

Quantity	Reported result	Interpretation
Validation	LRA and hard-wall HDM match OpenSANS S-matrix spectra within about 0.08% in the reported metrics	The discretization reproduces established spherical solutions
Sphere set	Sodium radii $R = 1\text{--}5\text{ nm}$; $\mu_p = 6.04\text{ eV}$	Simple-metal test bed for nonlocal and diffuse-surface effects
$R = 2\text{ nm}$, LRA	Dipole $= 3.49\text{ eV}$	Sharp-interface local baseline
$R = 2\text{ nm}$, HDM	Dipole $= 3.64\text{ eV}$	Hard-wall compressibility pushes screening inward and blueshifts
$R = 2\text{ nm}$, SC-HDM	Dipole $= 3.30\text{ eV}$	Diffuse density and induced charge spill outward and redshift
Surface mode	Bennett-type feature near 4.1 eV	Finite-width surface supports sign-changing multipole charge
Runtime	85.69 s LRA; 87.95 s HDM; 94.36 s SC-HDM	About 10% SC-HDM overhead for this spherical

MESOSCOPIC OPTICAL RESPONSE

STRUCTURED REVIEW - 2026

PHYSICAL COMPARISON AT $R = 2\text{ NM}$

LRA $3.49\text{ eV} \rightarrow$ hard-wall HDM $3.64\text{ eV} \rightarrow$ SC-HDM 3.30 eV , with a distinct Bennett surface feature near 4.1 eV .

6.4 Extraction of surface response

The paper evaluates a spherical charge-centroid quantity analogous to d_L and finds a Lorentzian feature at the Bennett resonance. This supports a multiscale route in which SC-HDM generates an interface response that is later used in a cheaper surface-corrected Maxwell solver. The comparison with a planar TDDFT fit is exploratory, however: a finite sphere is not a planar half-space and spatial dispersion $d(q, \omega)$ is omitted.

The SC-HDM response also develops a second zero near 4.62 eV and additional high-energy features whose number depends on the allowed spill-out extent. The authors identify this as a high-energy instability or truncation sensitivity of the selected functional and density-tail treatment. The extracted curve should therefore not yet be used as a universal sodium d parameter.

6.5 Contribution to the field

Established by the paper	Still open
Symmetry-adapted VIE reproduces spherical LRA/HDM benchmarks	Arbitrary 3D shapes, substrates, edges, and touching junctions
SC-HDM can approach LRA runtime for tested spheres	Whether similar efficiency survives without spherical symmetry
Diffuse-interface shifts and a Bennett mode are recovered	Direct experimental validation of the predicted sodium spectra
Finite-sphere surface response can be generated cheaply	Stable, causal, transferable planar $d(q, \omega)$ for real noble-metal interfaces
Framework accommodates alternative energy functionals	Interband realism, tunnelling, and systematic functional calibration

The conceptual arc from the 2021 equilibrium-profile paper to the 2026 VIE paper is therefore coherent. The first identifies how diffuse equilibrium structure contributes to surface response; the second makes a self-consistent continuum calculation of that structure and its dynamics far cheaper in a canonical geometry.

Selected references I
